A Sample Book Using Quarto with Standard LaTeX Book Class

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2023-06-23

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Chapter 1

Introduction

This file is a sample book demonstrating Quarto's capabilities with the standard LaTeX `book` document class. It supports PDF (via LaTeX) output. It is assumed that the reader has a basic working knowledge of stochastic calculus Karatzas and Shreve [1] and stochastic control theory Yong and Zhou [2].

Theorem 1.0.1 (Residue Theorem)**.** *Let f be analytic in a simply connected domain D. Then for any closed contour $\gamma \subset D$,*

$$\oint_\gamma f(z)\,dz = 2\pi i \sum_k \mathrm{Res}(f, z_k)$$

This template supports standard Quarto cross-references. You can refer to Theorem Theorem 1.0.1 in the text.

Chapter 2

Markdown Basics

The template is built on Quarto and Pandoc, leveraging standard LaTeX tools for PDF output.

2.1 Equations

Inline math: $E = mc^2$. Display math:

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u$$

2.2 Theorems and Proofs

Lemma 2.2.1 (Key Lemma)**.** *If f is continuous on $[a, b]$, then f is integrable on $[a, b]$.*

Corollary 2.2.1 (Easy Corollary)**.** *Every continuous function on a closed interval is bounded.*

Definition 2.2.1 (Model Definition)**.** A stochastic process $\{X_t\}_{t\geq 0}$ is a collection of random variables indexed by time.

2.3 Lists

1. First item
2. Second item
3. Third item

- Bullet one
- Bullet two

2.4 Tables

Table 2.1: A sample table

Variable	Description	Unit
x	Position	m
v	Velocity	m/s
a	Acceleration	m/s^2

Chapter 3

Bibliography

This template uses `natbib` with the `plainnat-doi` bibliography style for author-year citations Karatzas and Shreve [1].

Acknowledgment

This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this book.

Use the `[]{.appendix options="An Appendix"}` markup if you have a single appendix. Do not use `\section{}` after `\appendix` in LaTeX.

References

[1] Ioannis Karatzas and Steven E. Shreve. *Brownian Motion and Stochastic Calculus*. Springer New York, 1991.

[2] Jiongmin Yong and Xunyu Zhou. *Stochastic Controls: Hamiltonian Systems and HJB Equations*. Stochastic Modelling and Applied Probability. Springer New York, 1999.

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